Sabancı Universitesi **Syllabus Application**

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CS 301
Algorithms

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 301 - Algorithms

Time/Place

Time	Week Day	Place	Date
16:40-17:30	Wed	FENS-G077	Feb 16-May 22, 2026
14:40-16:30	Fri	FENS-G077	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Basic:1, Engineering:5

Prerequisites (MATH 204 and CS 300)

Corequisites CS 301R

Course Type Lecture

Program Requirements >

Instructor(s) Information

Hüsnü Yenigün

Email: yenigun@sabanciuniv.edu

Office Hours (by appointment): Tuesday 18:30-20:30

Please reserve a slot before 23:59PM of the previous day, latest.

Course Information

Catalog Course Description

This course will cover algorithms for a variety of problems, as well as general algorithm design and analysis techniques such as divide-and-conquer, dynamic programming, and greedy algorithms. Specific topics include algorithm analysis, recurrences and asymptotic analysis, searching, sorting, order-statistics, shortest path problems, and network-flows. An introduction to the computational complexity classes (such as P, NP, NP- hard, NP-complete, PSPACE) together with approximation algorithms and the performance evaluation of algorithm implementations in practice are also covered in the course.

Course Learning Outcomes:

1.	The students are expected to be able to analyse algorithms' resource usage by using asymptotic analysis and asymptotic notations.
2.	The students are expected to be able identify the algorithm design technique used by a given algorithm such as divide-and-conquer, dynamic programming, greedy, etc.).
3.	The students are expected to be able to use these algorithm design techniques to develop new algorithms for simple computational problems.
4.	The students are expected to be aware of some common computational problems and some known algorithms for these problems.
5.	The students are expected to know that there are limits of algorithmic approaches to the computational problems.
6.	The students are expected to be able to design tests to analyse the correctness and the performance of practical implementations of algorithms.

Course Objective

To prepare students 1) to analyze an algorithm's performance by asymptotic analysis methods, 2) to understand the role of data structures and programming paradigms on the performance of algorithms, and 3) to design efficient algorithms taking into account these important factors.
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Course Materials

Resources:

TEXTBOOK:
Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein

Lecture slides will be provided.

Technology Requirements:

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	40%	
Midterm	40%	1
Assignment	10%	3
Group Project	10%	1

Midterm Exam Date: April 5, 2026 @ 15:00-17:00
Final Exam Date: TBA

Similar to a failing threshold (which is decided at the end of the semester) for the overall numeric grade of the course, there will also be a failing threshold for the all the grading items (midterm, final, homeworks, project). If your grade from a grading item falls below the failing grade of a grading item, this will be a reason for failing the course. Again, similar to the failing threshold for the overall numeric grade of the course, failing thresholds for the midterm, final, homeworks, projects will be decided at the end of the semester.

Exam Schedule:

Time	Event	Details
Apr 5, 2026 15:00	Midterm: CS301 Midterm	

Policies

Course Policies:

Make-up Policy: If you miss exactly one of the midterm or final exam, and if you have a valid excuse (e.g. a medical condition, an official university event participation, etc.), then you can take the make-up exam. In this case, the grade of the make-up exam counted as the grade of your missing exam. The make-up exam can be an oral exam, a written exam, or both.

AI Policies

Guidelines on the "Use of Generative AI" for students

SU Academic Integrity Statement:

https://www.sabanciuniv.edu/en/academic-integrity-statement

Health Report Requirements

Student Medical Reports Instruction Letter

Weekly Content

Week	Topic
Week 1	Introduction, Algorithm Design Techniques, Growth of Functions
Week 2	Background, Recurrences, Substitution Method, Iteration Method, Master Method, Lower Bounds, Sorting in Linear Time
Week 3	Stability of Sorting Algorithms, Radix Sort, Medians and Order Statistics, Dynamic Sets on Binary Search Trees
Week 4	Dynamic Sets, on Binary Search Trees, Red-Black Trees
Week 5	Augmenting Data Structures, Dynamic Programming
Week 6	Dynamic Programming, Greedy Algorithms
Week 7	Amortized Analysis, Graphs
Week 8	Minimum Spanning Tree, Shortest Path Problems

Week	Topic
Week 9	NP-Completeness, Test Design (Functional and Performance Tests)
Week 10	Approximation Algorithms, Flow Networks
Week 11	Maximum Bipartite Matching, Sorting Networks
Week 12	Computational Geometry
Week 13	Randomized Algorithms
Week 14	coNP and PSPACE Complexity Classes